

DOCUMENT NAME

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1. User Stories

1.1. Admin Authentication & Account

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
A01	Admin	Log into the admin website	I can access the management dashboard	Valid email + password → redirect to Admin Dashboard; invalid credentials → clear error message
A02	Admin	Log out from the system	I can secure my admin session	Session/token is invalidated and admin is redirected to Login page
A03	Admin	Reset my password	I can recover access if I forget it	Reset via email/OTP, password must be updated and old token becomes invalid

1.2. User & Caregiver Management

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
A04	Admin	View list of all users and caregivers	I can monitor all registered accounts	Paginated, sortable table with filters (role, status)
A05	Admin	Search users by name, email, or ID	I can find specific accounts quickly	Only matching users are displayed; search is case-insensitive
A06	Admin	View detailed profile of a user	I can understand their devices, alerts, and status	Detail view shows personal info, assigned devices, last login, recent alerts
A07	Admin	Edit user profile information	I can correct or update wrong data	Changes are stored and immediately reflected in subsequent views
A08	Admin	Deactivate or delete a user account	I can remove fake or inactive accounts	Confirm dialog; deactivated users cannot log in; deleted users are hidden from default list but kept in audit log

1.3. Device & Registration Management

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
A09	Admin	View list of all registered devices	I can monitor device inventory and status	Table shows type, owner, online/offline, last heartbeat
A10	Admin	Review and approve new device registration requests	I can ensure only trusted devices are connected	Pending list; approve/deny; owner receives notification
A11	Admin	Reassign a device to another user	I can handle changes in ownership or caregiver	Reassignment updates mapping and appears in both user and device views
A12	Admin	Disable a malfunctioning device	I can prevent false or spam alerts	Disabled devices are not allowed to send alerts; status clearly shown

1.4. Monitoring, Alerts & Health Overview

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
A13	Admin	View live monitoring dashboards	I can see current status of multiple elderly users	Dashboard cards show fall status, last sensor data, device online status
A14	Admin	View fall incident history by user or device	I can investigate frequent or critical incidents	Filter by date range/user/device; newest first; includes severity and location
A15	Admin	View health summary (BMI, blood pressure, blood sugar) trends	I can identify high-risk users	Charts per user with configurable time range and thresholds

1.5. Reporting, Logs & System Settings

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
A16	Admin	Generate weekly/monthly system usage reports	I can analyze adoption and reliability	Report includes #active users, #alerts, uptime; downloadable as PDF/Excel
A17	Admin	View system activity logs	I can audit critical actions and errors	Logs show timestamp, actor, action, result; searchable by user and type
A18	Admin	Manage notification templates	I can customize alert messages (email, push, Telegram)	Template fields can be edited and previewed; new alerts use latest template

A19	Admin	Configure global alert thresholds and rules	I can tune the sensitivity of fall detection and reminders	Changes stored centrally; rules applied to new events immediately
A20	Admin	Manage admin roles and permissions	I can limit what each admin can do	Roles (e.g., Super Admin, Support) with defined permissions; enforcement on UI and backend

2. User stories for Personal User / Caregiver

2.1. Authentication & Profile

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U01	Personal user	Register for an account	I can start using the system	Email is unique; strong password rules; verification email is sent
U02	Personal user	Verify my account via email/OTP	I can activate my account securely	After verification, user can log in; incorrect code shows error
U03	Personal user	Log into my account	I can access my dashboard	Valid credentials → Dashboard; invalid → error message
U04	Personal user	Reset my password	I can recover my access	Reset link or OTP is sent; password updated correctly
U05	Personal user	Log out from the system	I can secure my information	Token cleared; redirected to Login
U06	Personal user	Edit my profile	I can keep my personal and medical info up to date	Name, phone, address, date of birth, emergency contact editable
U07	Personal user	Upload a profile photo	I can personalize my account	Valid image formats; displayed immediately after upload

2.2. Device Pairing & Status

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U08	Personal user	Register and pair my IoT device (camera/ESP32)	I can receive sensor and fall data	Pairing wizard; successful pairing shows device in Device list
U09	Personal user	See connection status of my devices	I can know if monitoring is working	Status icon: online / offline / error, updated in near real time
U10	Personal user	Manually disconnect my device	I can stop monitoring when not needed	Disconnect stops incoming data; UI shows “disconnected”
U11	Personal user	View battery status and health of my devices	I can charge or replace them in time	Battery percentage shown; warning under 20%

2.3. Monitoring & Fall Alerts

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U12	Personal user	View my live sensor data	I can monitor my current physical condition	Data auto-refresh (e.g., every 2 seconds) while screen is open
U13	Personal user	View fall detection alerts in real time	I can react immediately to dangerous events	Red alert card with timestamp & device; visible on dashboard
U14	Personal user	Test the fall alert manually	I can verify that the system is working	Pressing “Test fall” triggers a simulated alert and notification
U15	Personal user	View history of all fall incidents	I can review previous accidents	List sorted by newest first, filter by date; can open detail view
U16	Personal user	See the location of the last fall on a map	I can understand where it happened	Map marker with time; address text shown below map
U17	Caregiver	Receive push notification when a fall occurs	I can support the elderly quickly	Notification sent to linked accounts within a few seconds

2.4. Health Tracking & Reminders

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U18	Personal user	Record and view my weight & BMI	I can track changes in body condition	Can add manual entry; chart updated; BMI auto-calculated
U19	Personal user	Record and view my blood pressure	I can monitor cardiovascular health	Systolic/diastolic/pulse stored with timestamp; highlight critical values
U20	Personal user	Record and view my blood sugar	I can manage diabetes risk	Fasting/post-meal type; trend chart; high/low flags
U21	Personal user	Receive reminders to drink water	I can maintain good hydration	Schedule configurable; notification shown at each reminder time
U22	Personal user	Receive reminders to move when inactive	I can avoid sitting too long	Inactivity threshold (e.g., 60 minutes) configurable; alert message displayed
U23	Personal user	Receive weekly summary of my activities and health metrics	I can see my progress over time	Weekly email/app report with steps, falls, vital stats, trends

2.5. Doctor AI, Medical Contact & Support

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U24	Personal user	Ask health-related questions to an AI doctor	I can get quick basic advice	Chat UI; responses must include disclaimer “not a replacement for doctor”
U25	Personal user	Store emergency medical contacts	I can contact them quickly during alerts	At least one contact required; phone/email stored and visible
U26	Caregiver	Receive health reports shared by the elderly user	I can follow their condition easily	Share via email/QR; link opens a read-only report page
U27	Personal user	Access a help & FAQ page	I can learn how to use the system	Contains tutorials, FAQs, contact support links

2.6. Settings & Appearance

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
U28	Personal user	Configure notification preferences (channels & severity)	I can avoid being overwhelmed by alerts	User can turn on/off email, push, Telegram; settings are persisted
U29	Personal user	Switch between light and dark mode	I can reduce eye strain	Theme is toggled instantly and preference stored locally/account-wide
U30	Personal user	Delete my account	I can leave the system permanently	Confirm dialog with warning; after deletion, login impossible and data anonymized where required

3. System / Backend / IoT Stories

3.1. Security & Identity

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
S01	Backend	Validate and authenticate user credentials	I can allow secure login	Returns JWT/secure token for valid credentials; rejects invalid attempts
S02	Backend	Hash and store user passwords securely	I can protect user data	Uses strong hashing (e.g., bcrypt/Argon2); plain passwords never stored
S03	Backend	Manage user roles and permissions	I can enforce authorization	Role check before accessing any protected endpoint

3.2. Data Management & APIs

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
S04	Backend	Store all sensor, fall, and health data with timestamps	I can retrieve and analyze it later	Data persisted with user ID, device ID, and time; no silent failures
S05	Backend	Provide REST/GraphQL APIs for web and IoT clients	I can integrate multi-platform apps	Response time ≤ 500 ms for 95% common requests under normal load
S06	Backend	Log all critical system activities and errors	I can audit and debug the system	Logs include actor, action, result, and time; searchable by tool

3.3. Notification & Reliability

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
S07	Backend	Send push / email /Telegram notifications for fall events	I can alert stakeholders in real time	99% of alerts delivered within defined SLA (e.g., 3–5 seconds)
S08	Backend	Queue notifications When services are temporarily down	I can avoid losing alerts	Failed sends are retried until success or expiry time

3.4. IoT Device Behavior

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
S09	IoT device	Detect fall events using MPU6050 and/or camera	I can trigger alerts automatically	Tested accuracy $\geq$ target threshold (e.g., 80%) under test scenarios
S10	IoT device	Send sensor data to backend via Wi-Fi	I can keep data up to date	Data packet sent every N second; backend receives without duplication
S11	IoT device	Automatically reconnect after disconnection	I can maintain stable communication	Device attempts reconnect periodically; within 10s after network is back
S12	IoT device	Send battery and device health status	I can monitor hardware reliability	Low battery (<20%) and sensor errors flagged to backend

3.5. Privacy & Compliance

ID	As a/an	I want to ...	so that ...	Acceptance Criteria
S13	Backend	Store and process data according to privacy consent	I can respect user choices	Each user has consent flags; backend checks before sharing/exporting data
S14	Backend	Anonymize personal identifiers when deleting accounts	I can protect user privacy	After deletion, data used for statistics no longer contains direct identifiers

4. System Architecture

Reference to Architecture Document v1.2

5. Product backlog

Reference to Product Backlog Document v1.2

6. References

- [1] . Scrum Process: A gentle introduction <https://scrumguides.org/>
- [2]. Scrum Guide - <https://scrumguides.org/scrum-guide.html>
- [3]. [https://en.wikipedia.org/wiki/Scrum_\(software_development\)](https://en.wikipedia.org/wiki/Scrum_(software_development))